Reducing the Cogging Torque Effects in Hybrid Stepper Machines by Means of Resonant Controllers

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***Abstract*—Permanent magnet machines are not free from the interaction between magnets and the stator and rotor slots, which causes an undesired disturbing torque. Such cogging or detent torque is especially larger with salient pole machines, as it is the case of the permanent magnet hybrid stepper machines (PMHSM). Depending on the application requirements, these torque perturbations can be unacceptable and the application of solutions that minimizes the cogging torque effects are mandatory. This paper analyzes the minimization of the cogging torque using resonant controllers. More specifically, this paper details the analysis and design of a speed-adaptive resonant controller, which is not only directly designed in the Z-domain but also considers the current (or torque) inner-loop delay. Pole-zero placement and the disturbance-rejection-frequency response are attained in the design of the speed and position speed-adaptive controllers. Experimental results with two off-the-shelf PMHSMs demonstrate the superior performance of the proposal in both speed and position closed-loop applications for tracking, as well as in disturbance (load impact) rejection tests and against inertia variations. A comparison with a conventional PI is carried out from the design stage to experimental results and the improvement of the proposal is numerically quantified.**

***Index Terms*—Cogging torque, field-oriented control, permanent magnet hybrid stepper machine, position control, resonant controller, speed control.**

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I. Introduction

NOWADAYS, permanent magnet hybrid stepper machines (PMHSMs), also known as stepper motors (SMs), are considered as an attractive technology for position-controlled motion applications [1]–[3]. Hence, the PMHSM's presence in the market share is remarkable [4]–[7], as well as their role in numerous state-of-the-art applications. Among these endless list of applications, several works could be mentioned, like [8], where the SM is a satisfactory choice for driving the control rods of a modular high-temperature gas-cooled reactor. Also in [9], SMs are used to position each of the 88 intersection cells of the three undulator segments of the European X-ray free electron laser. Likewise, SMs are the choice in [10] to mechanically reconfigure a novel antenna, in [11] to actuate in a spectrophotometer, in [12] to track the sun, in [13] to drive a vertical column, and in [14] as a disk driver.

PMHSMs most relevant feature is the capability of performing accurate position control without the use of any electromechanical feedback sensor by means of the well-known micro-stepping technique [15]–[17]. However, such open-loop positioning technique suffers from two important inconveniences, which are, the always threatening loss of steps [18], [19] with the consequent position error, and the use of maximum current [20], which implies poor efficiency, as well as extra heating. Both of the previous inconveniences are skipped by using vector control (also known as field-oriented control) with an encoder or resolver feedback, similarly as other servo applications [21], [22]. However, when using vector control at low speeds or standstill in position control applications, the undesired cogging torque effects appear.

Cogging torque is a distorting periodic torque generated by the tendency of the rotor magnetic field to be aligned with the stator poles [23], [24]. These torque oscillations produce vibrations and additional noise and are present even if no torque production is commanded (i.e., without currents circulating through the stator). Such distorting torque is position dependent and its periodicity per revolution depends on the number of magnetic poles and the number of teeth on the stator. Therefore, the cogging torque can be treated as a periodic disturbance, whose amplitude value is almost constant, but its frequency is a multiple of the mechanical frequency of rotation.

In servo drive applications, where precise speed or position control is required, such cogging torque is almost imperceptible at medium and high speeds [25]. Some research papers claim that these values are relatively negligible as compared to standard electromagnetic torques, so they can be neglected [25], [26]. However, at low speeds it can create high ripples in the mechanical speed and extra difficulties to achieve the targeted position. In many applications, such effects are unaffordable and its minimization becomes mandatory [27].

A great amount of scientific literature regarding cogging torque minimization can be found, which can be broadly divided into two approaches: 1) the ones focused on altering the physical machine design [24], [28]–[34]; and 2) the ones based on precommissioning processes to evaluate the cogging torque for further compensation [27], [35], [36]. Inconveniences exist in both approaches, the most relevant being the unavoidable actions to be taken before the manufacturing for the first approaches and the accurate precommissioning process, with the compulsory use of high-resolution resolvers or encoders at some point for the second ones.

With the aim of opening a third approach to tackle the cogging torque, this paper proposes the use of resonant controllers. Such controllers are based on the application of the internal model principle [37] and, consequently, they constitute an adequate approach to the problem of tracking periodic references or, as in this case with the cogging torque, rejecting periodic disturbances with a finite number of significant harmonics. Resonant controllers [38]–[42] have been already applied, among others, to mechanical systems[43] and power electronic inverters [39], [41], [42]. In this paper, however, there is the particular scenario of needing to adapt its resonant frequency as a function of the motor mechanical speed.

This paper details the analysis and design of a speed-adaptive resonance controller, which not only is directly designed in Z-domain but also considers the current (or torque) inner-loop delay. Pole-zero placement and disturbance-rejection-frequency response have been attained in the design. In order to evaluate the superiority of the proposed resonant speed-adaptive controller, a comparison with a standard proportional and integral (PI) is carried out. Finally, experimental results with two different off-the-shelf PMHSMs demonstrate the superior performance of the proposal in both speed and position tracking closed-loop applications, as well as in disturbance rejection (load impact) tests. Moreover, a higher robustness against inertia variation is experimentally corroborated.

II. PMHSM Model Including Cogging Torque

The α/β two-phase bipolar PMHSM electrical equations are given in (1) and (2) [44].

$$L\frac{di_\alpha}{dt} = v_\alpha - Ri_\alpha - \Phi_{\mathrm{PM}} N_r \omega_m \sin\left(N_r \omega_m t\right) \tag{1}$$

$$L\frac{di_\beta}{dt} = v_\beta - Ri_\beta - \Phi_{\mathrm{PM}} N_r \omega_m \sin\left(N_r \omega_m t - \frac{\pi}{2}\right) \tag{2}$$

where v_α, v_β, i_α, and i_β are the stator voltages and currents; R and L are the stator nominal resistance and inductance; Φ_{PM} is the PM flux; ω_m is the mechanical angular speed; and N_r is the PMHSM rotor teeth-number (which effectively is equivalent to the pole-pair number as in other ac machines, in the sense that it is the factor between electrical and mechanical angular speeds). The electromagnetic torque (τ) produced by the PMHSM is the interaction (or vector product) between all magnetic fluxes and currents, as follows:

$$\tau = N_r \Phi_{\mathrm{PM}} \sin\left(N_r \omega_m t + \frac{\pi}{2}\right) i_\beta - N_r \Phi_{\mathrm{PM}} \sin\left(N_r \omega_m t\right) i_\alpha. \tag{3}$$

The cogging torque (τ_c) should be mathematically represented in terms of the reluctance change between the coils and magnets [28]. However, attending just at its impact in the speed loop, the following expression is preferred:

$$\tau_c = \sum_{j=1}^{\infty} K_c^j \sin\left(j\omega_e t + \phi^j\right) \tag{4}$$

where $\omega_e = N_r \omega_m$ is the electrical angular speed, while K_c^j and ϕ^j stand for the amplitudes and phases respectively, of the different harmonic components. In this paper and with the PMHSMs used, the most remarkable (and almost unique) component of the cogging torque is for $j = 1$. It must be pointed out that, while other approaches such as [36] require the exact knowledge of the parameters K_c^j and ϕ^j to fight against the cogging, the resonant based controller presented in this paper does not. Nevertheless, it is worth mentioning that the most relevant value has been found to be equal to $K_c^1 = 0.175$ N·m for the PMHSM further detailed in Table I.

Finally, the mechanical speed is modeled in (5) as a first-order system

$$J\frac{d\omega_m}{dt} = \tau - \tau_c - \tau_L - B\omega_m \tag{5}$$

where B and J are the friction and inertia, (τ_L) the external load torque, and (τ) and (τ_c) the previously defined electromagnetic and cogging torques, respectively.

III. Torque Control Loop

Despite the fact that field-oriented control, though implemented here, is out of the scope of this paper, it has been found that any influence in the speed loop must be carefully considered, and therefore, modelled in order to further face the

TABLE I
Parameters of the Test Bench

PMHSM (reference SY57STH76-2804A)		
Parameter	**Value**	**Units**
Step angle / Phase no. / N_r	1.8 / 2 / 50	° / — / —
R / L / Voltage constant (KE)	1.13 / 3.6 / 0.524	Ω / mH / V·s·rad^{-1}
Holding / Detent Torques	1.85 / 0.067	Nm / Nm
Test bench		
Parameter	**Value**	**Units**
Total (*) Inertia / Friction	$0.3 \cdot 10^{-3}$ / $12.5 \cdot 10^{-3}$	kg·m^2 / Nm·s·rad^{-1}
Sampling period (T)	$500 \cdot 10^{-6}$	s
Encoder pulses / Speed resol.	4·2500 / 12	counts·rev^{-1} / rpm

(*) Inertia and friction include the mechanical coupling and the dc machine, i.e., are the total rig inertia and friction values.

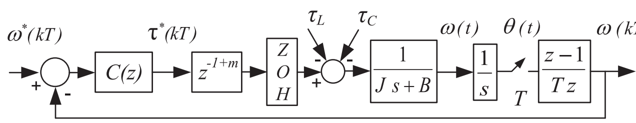

Fig. 1. Speed control loop.

speed controller design with warranties. In this sense, it must be pointed out that the torque dynamics correspond to a second-order system with a tiny overshoot and a settling time at two percent equal to the speed loop sampling time (T). Also, it must be taken into account that the current controllers [44] have been implemented with a sampling rate ten times faster than the speed loop. Therefore, an accurate torque loop model is a portion (m) of one speed loop sample time delay ($z^{(-1+m)}$), as it is indicated in Fig. 1. Since the Z-transform is defined for an entire number of delays and not for a fractional portion of them, the modified Z-transform (Z_m) [45] needs to be used instead.

IV. Speed Control Loop Including the Proposed Resonant Controller

Among all the speed control loop blocks, Fig. 1 contains the mechanical first-order plant in series with an integrator, which is mathematically expressed as

$$G'(s) = \frac{1/B}{(J/B)s + 1}\,\frac{1}{s}. \tag{6}$$

Considering the data-sampled nature of the controller, the well-known zero-order hold transformation method [45], expressed in (7), has been used to obtain the plant transfer function (8) in Z-domain.

$$G'(z) = \left(1 - z^{-1}\right) Z\left\{\frac{G'(s)}{s}\right\} \tag{7}$$

$$G'(z) = \frac{1}{B\mathrm{a}}\,\frac{z\left(\mathrm{a}T - 1 + e^{-\mathrm{a}T}\right) + \left(1 - e^{-\mathrm{a}T} - \mathrm{a}Te^{-\mathrm{a}T}\right)}{(z-1)(z - e^{-\mathrm{a}T})} \tag{8}$$

where $\mathrm{a} = B/J$.

On the other hand and as justified in Section III, once the inner torque loop is considered as a fractional delay, the modified Z-transform must be applied as stated in (9) and then (10) is obtained.

$$G_m(z) = \left(1 - z^{-1}\right) Z_m\left\{\frac{G'(s)}{s}\right\} \tag{9}$$

$$G_m(z) = \frac{1}{B\mathrm{a}}\,\frac{z^2x_2 + zx_1 + x_0}{z\,(z-1)\left(z - e^{-\mathrm{a}T}\right)} \tag{10}$$

where

$x_2 = \mathrm{a}mT - 1 + e^{-\mathrm{a}mT}$

$x_1 = 1 + \mathrm{a}T(1-m) + e^{-\mathrm{a}T}(1 - \mathrm{a}mT) - 2e^{-\mathrm{a}mT}$

$x_0 = e^{-\mathrm{a}T}(-1 + \mathrm{a}T(-1+m) + e^{-\mathrm{a}mT})$.

The measured speed is obtained by the discrete-time derivation of the encoder measured position [21], whose transfer function is the last one on the right of Fig. 1. The resultant $G(z)$ plant, whose transfer function is expressed in (11), contains three poles, two of them at the origin (labeled p_3, p_4 in Fig. 2),

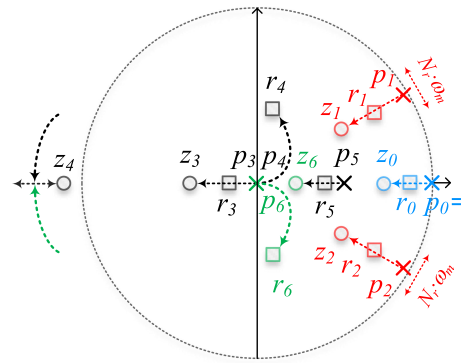

Fig. 2. Pole-zero map in Z-domain and root-locus evolution of the speed loop.

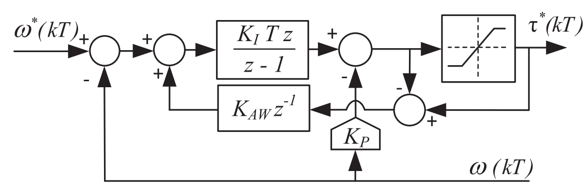

Fig. 3. Conventional integral proportional speed controller with the required antiwindup.

a third pole (labeled p_5), whose value is equal to $e^{-\mathrm{a}T}$, and finally, two finite zeros (labeled z_3 and z_4). Specific values are fixed by the mechanical parameters, whose data is detailed in Table I

$$G(z) = G_m(z)\frac{z-1}{Tz} = \frac{1}{B\mathrm{a}T}\,\frac{z^2x_2 + zx_1 + x_0}{z^2(z - e^{-\mathrm{a}T})}. \tag{11}$$

A. Conventional PI Controller

The PI actions combined are one of the most widespread speed controllers for electric machine commercial drives [21], [46]. In this paper, the PI is tuned using (12) to fulfil the specifications of settling time at 2% ($ST_{2\%}$) and damping factor (ζ), which is fixed to 1 in order to avoid any overshoot [44].

$$K_I = \frac{5.8^2 J}{\zeta^2 ST_{2\%}^2}\,; \qquad K_P = \frac{5.8J}{ST_{2\%}} - B. \tag{12}$$

In order to implement it digitally, the traditional approach, valid for the majority of electric machine drives [21], [44], consists of using the backward Euler rectangular approximation

$$\frac{1}{s} \equiv T\frac{z}{z-1}. \tag{13}$$

In this research, the integral proportional (IP) version has been implemented instead of the conventional PI and the effect of the additional zero has been avoided without the need of a prefilter [44], as it is shown in Fig. 3.

B. Proposed Resonant Integral (RI) Controller

In order to guarantee the regulation capability and the minimization of the cogging torque effect, integral $I(z)$ and resonant

$R(z)$ actions given in (14) and (15), are initially proposed

$$I(z) = \frac{1}{z-1}. \tag{14}$$

The specifications of the resonant controller will be fixed by its poles and zeros, (labeled p_1, p_2, and z_1, z_2, respectively), indicated in (16) and (17) (and printed in red color in Fig. 2). Equivalently, its damping factors (ζ_z, ζ_p) and natural frequencies (ω_z, ω_p) define their behaviour. Actually, the ratio (ζ_p/ζ_z) determines the peak of attenuation, while the damping (ζ_z) fixes the width [21]

$$R(z) = \frac{1-c+d}{1-a+b}\,\frac{z^2-az+b}{z^2-cz+d} \tag{15}$$

where, $a = 2e^{-T\zeta_z\omega_z}\cos\left(T\omega_z\sqrt{1-\zeta_z^2}\right)$, $b = e^{-2T\zeta_z\omega_z}$, $c = 2e^{-T\zeta_p\omega_p}\cos\left(T\omega_p\sqrt{1-\zeta_p^2}\right)$, and $d = e^{-2T\zeta_p\omega_p}$.

$$p_{1/2} = e^{-T\zeta_p\omega_p}\left[\cos\left(T\omega_p\sqrt{1-\zeta_p^2}\right) \pm j\sin\left(T\omega_p\sqrt{1-\zeta_p^2}\right)\right] \tag{16}$$

$$z_{1/2} = e^{-T\zeta_z\omega_z}\left[\cos\left(T\omega_z\sqrt{1-\zeta_z^2}\right) \pm j\sin\left(T\omega_z\sqrt{1-\zeta_z^2}\right)\right]. \tag{17}$$

As commonly done, the use of the phase-lead controller given in (18) has found to be of great help to enlarge all stability margins, and therefore, it has been added

$$\mathrm{PL}(z) = \frac{z-z_6}{z(1-z_6)}. \tag{18}$$

Since the relative degree of the initially proposed controller is 1, an additional zero (z_0) is included in the final biproper controller transfer function $C(z)$ defined in (19), which also includes a gain K. Such additional zero will have a great influence in the slowest closed-loop pole location, and therefore, it will fix the speed of closed-loop dynamics. However, such zero might worsen the transient closed-loop response, and this is the reason for including the unity-gain prefilter given in (20). Such prefilter not only is designed to cancel the zero (z_0), but also to introduce a zero at the origin, since it aids to damp and stabilize the speed and position closed-loops, respectively

$$C(z) = \left(\frac{z-z_6}{z(1-z_6)}\right)K\left(\frac{z-z_0}{z-1}\right)\left(\frac{1-c+d}{1-a+b}\right)\left(\frac{z^2-az+b}{z^2-cz+d}\right) \tag{19}$$

$$\mathrm{PF}(z) = \frac{z(1-z_0)}{z-z_0}. \tag{20}$$

The open-loop transfer function is detailed in (21) and its pole-zero map, together with the root locus, are illustrated in Fig. 2, where the closed-loop poles are labelled as $r_0, r_1, r_2, r_3, r_4, r_5$, and r_6

$$\begin{aligned} L(z) &= C(z)\cdot G(z) \\ &= K'\frac{z-z_3}{p_3p_4(z-p_5)}\,\frac{z-z_6}{z(1-z_6)}\,\frac{z-z_0}{z-1}\,\frac{z-Rz_{1/2}\pm jIz_{1/2}}{z-Rp_{1/2}\pm jIp_{1/2}}. \end{aligned} \tag{21}$$

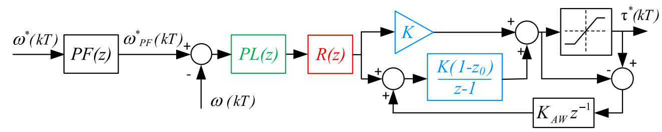

Fig. 4. Proposed speed controller with the prefilter (PF), the phase-lead (PL), the resonant (R), and the separated integral part with its antiwindup.

A saturation in the reference torque is unavoidable, and therefore, an antiwindup in the integral part is also required. The final implemented controller responds to the structure mathematically, detailed in (22) and schematically represented in Fig. 4. With this structure, it has been found that the integral part has no numerical errors since the coefficient 1 of the integrator is not further multiplied by any other number, and therefore, not numerically altered

$$C(z) = \frac{z-z_6}{z(1-z_6)}\,\frac{1-c+d}{1-a+b}\,\frac{z^2-az+b}{z^2-cz+d}\,K\left(\frac{1-z_0}{z-1}+1\right). \tag{22}$$

It must be pointed out that $R(z)$, defined in (15), is adaptive and its four coefficients (a, b, c, and d) need to be updated according to

$$\omega_p = \omega_z = \frac{1\cdot 50\cdot \omega_{\mathrm{PF}}^*(kT)}{\sqrt{1-2\zeta_p^2}} \tag{23}$$

where factors 1 and 50 are justified in (4) and given in Table I, respectively. Despite the fact that the natural frequency (ω_p) is not the resonant frequency ($\omega_r = \omega_p\sqrt{1-2\zeta_p^2}$) [45], given the small value of the selected damping factor for the poles ($\zeta_p = 0.01$), they could be considered to be the same.

C. RI and PI Tuning and Discussion

The accurate tuning of the speed controller strongly depends on the mechanical plant parameters, where the most relevant ones are listed in Table I.

The sensitivity transfer function has a paramount importance when evaluating the effectiveness of the cogging torque rejection. In order to evaluate the superior performance of the proposed RI controller, a comparison with the well-established PI controller has been carried out all over the investigation, from the early design up to the last experimental result. In order to achieve a "fair comparison," the criteria of having the same sensitivity transfer function (24) at low frequencies has been adopted when tuning both controllers

$$\frac{\omega(z)}{\tau_c(z)} = \frac{G'(z)(z-1)/(Tz)}{1+L(z)}. \tag{24}$$

These equivalences are infinite and among them, the values indicated in Table II have been considered for this case study. The resultant sensitivity Bodes are compared in Fig. 5, where the frequency responses are almost the same but in the resonant peak, which needs to vary according to the frequency component of the cogging torque. Fig. 5 corresponds to a mechanical speed equal to 6 r/min, which implies (considering the factor 50) that

TABLE II
RI AND PI TUNING PARAMETERS

Controller	Parameters
RI	z_6=0.7 z_0=0.98 K= 0.03 ζ_p=0.01 ζ_z=0.9 $\omega_p = \omega_z = \frac{\omega^*_{PF}(kT)\cdot 50}{\sqrt{1-2\zeta_p^2}}$
PI	$ST_{2\%}$ = 90 ms ζ=1

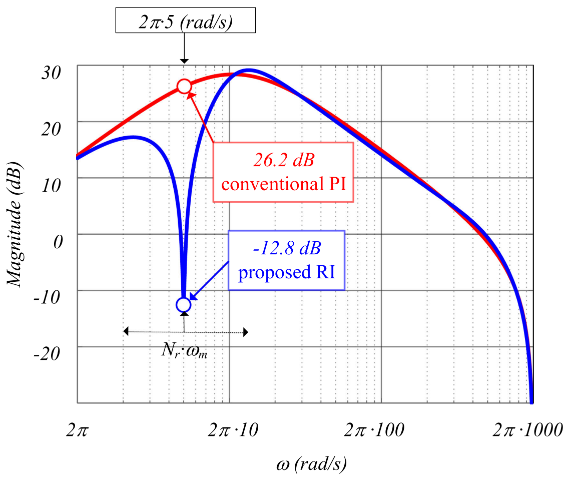

Fig. 5. Sensitivity Bode for conventional PI (red) and proposed RI (blue) when $n_m^* = 6$ r/min and the resonant attenuation peak is at 5 Hz.

the attenuation peak is placed at $2\pi\cdot 5$ rad/s with a gain of -12.8 dB; -39 dB less than the PI; which is imposed by the ratio (ζ_p/ζ_z) as follows:

$$20\log_{10}\frac{0.01}{0.9} = -39\,\text{dB}. \tag{25}$$

The adaptation of the resonant controller to track the cogging perturbation in accordance with the speed has a direct impact in all the pole-zero map. Therefore, not only the poles from the resonant controller vary, but also all six closed-loop poles do. Special care should be taken when dealing with the actual plant. In fact, the lower the inertia, the more difficult of the resonant tuning may become. In this sense, it must be pointed that:

1) reducing the overall controller RI gain K from (19) may eventually be useful;
2) at high speeds, the closed-loop poles r_4 and r_6 (see Fig. 2) travel very rapidly outside the unitary circle. However, at such speeds, the cogging torque is filtered by the mechanical plant itself and eventually its effects in the speed become unnoticeable. The adopted practical solution has been to adapt ω_p and ω_z up to a certain, experimentally found, speed threshold (around 150 r/min in this paper), after which the resonant controller is no longer updated. In other words, the adapting process works within the positive and negative speed threshold inside which the cogging torque becomes noticeable;
3) the introduction of the phase-lead controller defined in (18) enlarges such speed threshold. Outside this range, the proposed controller works with a fixed resonant, and therefore, equivalently as the PI.

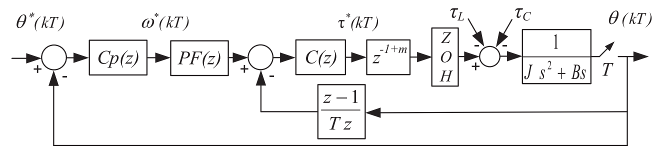

Fig. 6. Position control loop.

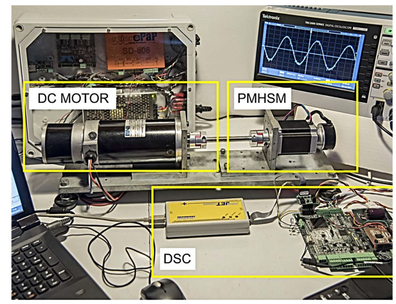

Fig. 7. Experimental rig composed by the PMHSM, dc MOTOR with its drive and the digital signal controller with the power converter on the right.

Finally, it should be remarked the importance of the branch defined by the zero z_0 and the integrator open-loop pole p_0, since such branch contains the slowest closed-loop pole r_0, which fixes the dynamics (see Fig. 2). Moreover, with the introduction of the prefilter, given in (20) and illustrated in Fig. 4, any influence of the nearby zero z_0 is totally canceled.

V. Position Control Loop

Fig. 6 illustrates the position loop, composed by the inner speed closed-loop and the $Cp(z)$ as a position controller, which is just a proportional gain for both (RI and PI) options. This position loop scheme has the measured sampled position in the feedback path and consequently it must be considered in the open-loop transfer function (26). With $Cp(z) = 2$, a good performance, in the double sense of achieving smooth motion during the tests and obtaining an illustrative comparison with both RI and PI controllers, has been achieved

$$Lp(z) = Cp(z)PF(z)\frac{(L(z)Tz)/(z-1)}{1+L(z)}. \tag{26}$$

VI. Experimental Results

Fig. 7 illustrates the four-quadrant experimental rig, which consists of a dc motor to emulate any type of load, the off-the-shelf PMHSM (reference SY57STH76-2804A [7]) and the Texas Instruments TMS320F28335 (150 MHz floating point) digital signal controller (DSC).

Reference tracking and disturbance rejection are the two main tests to evaluate the performance of any controller. Consequently, three test types have been carried out: load impact, speed, and position closed-loop controls. An encoder of 10 000 pulses per revolution has been used, which implies a speed resolution of 12 r/min, justified in (27). Such an inherent ripple

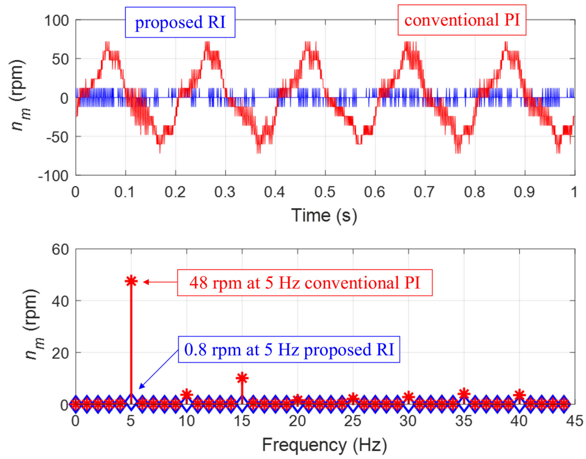

Fig. 8. Sinusoidal load impact disturbance rejection comparison for the proposed RI (blue) versus the conventional PI (red) and their FFTs.

appears all over the speed waveforms.

$$12\,(\text{r/min}) = \frac{\Delta\theta}{T} = \frac{1\,(\text{pulse})}{0.5\cdot 10^{-3}\,(s)}\,\frac{60\,(s)}{1\,(\text{min})}\,\frac{1\,(\text{rev})}{10^{4}\,(\text{pulse})}. \quad (27)$$

A. Load Impact

Instead of dealing with a step, as usually done, it has been considered much more appropriate to test the load impact against a periodic disturbance, since its rejection is the main goal of the RI controller. Hence, during such test, the dc motor is driven in closed-loop torque control with a sinusoidal reference of constant amplitude and constant frequency. The low speed scenario has been selected since it is when the unwanted cogging torque influence becomes stronger. Specifically, a frequency for the dc motor sinusoidal torque reference of 5 Hz has been chosen, which would correspond to the main cogging torque component when the machine would rotate as slow as 6 r/min ($fe = n_m(\text{r/min})\frac{N_r}{60}$). On the other hand, the PMHSM is in speed closed-loop control with a reference value equal to 0 r/min, and therefore, fighting to keep the rotor at zero speed. Fig. 8, which illustrates the mechanical speed n_m(r/min) and their fast fourier transform (FFTs) for the RI and PI controllers, shows an excellent and much superior performance of the former. Also in Fig. 8, the harmonic component attenuation at 5 Hz is reduced from 48 r/min to 0.8 r/min, which corresponds to −35.5 dB (where the theoretical value was −39 dB, as illustrated in Fig. 5).

With this test, the application of resonant controllers to mitigate periodic load impact perturbations under steady-state conditions has been positively demonstrated. Also, it has been experimentally corroborated that the reduction of the tuning factor ζ_p further attenuates the disturbance. However, extreme low values of ζ_p makes the resonant controller very effective at steady state, but less effective under speed control and position control where the speed varies.

B. Speed Loop

The lower the speed, the higher the challenge becomes to mitigate the unwanted speed oscillations due to the cogging

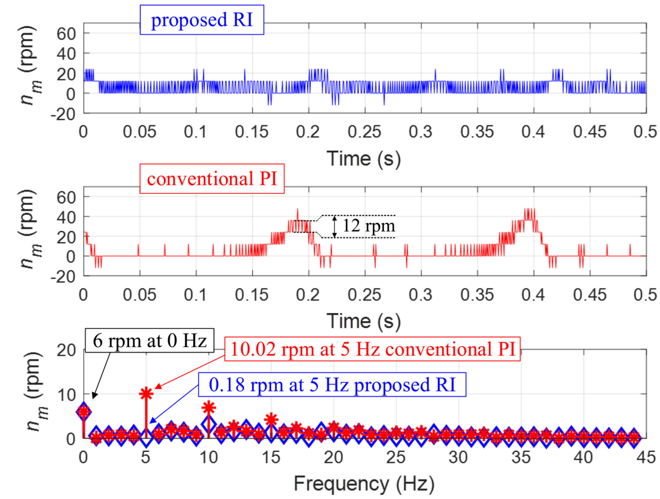

Fig. 9. Tracking 6 r/min comparison. Time waveforms and their FFTs (proposed RI in blue and conventional PI in red).

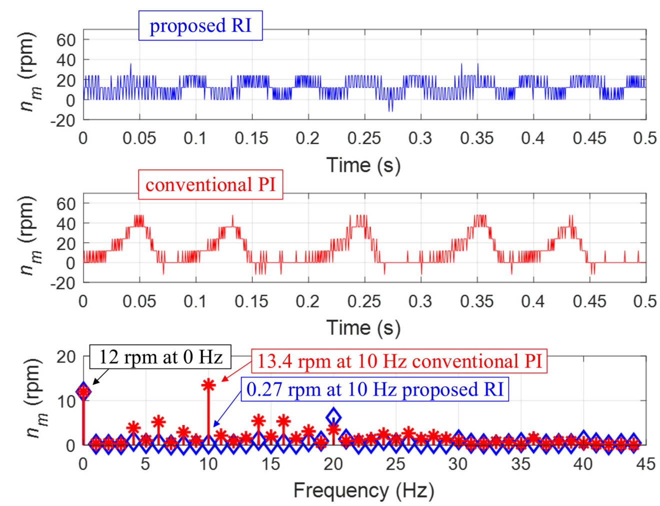

Fig. 10. Tracking 12 r/min comparison. Time waveforms and their FFTs (proposed RI in blue and conventional PI in red).

torque. Speeds of 6, 12, 18, and 24 r/min have been chosen in Figs. 9–12, because their first harmonic components lie in the exact values of 5, 10, 15, and 20 Hz, respectively. Hence, all FFTs shown are much cleaner and of easier comparison. It is relevant to point out that the steady state speed figure appears in the dc component of the FFTs. Table III shows the calculated total harmonic distortion (THD) for the first 44 components, as indicated in the following expression:

$$\text{THD} = \sum_{\text{Freq}=1(\text{Hz})}^{44(\text{Hz})} \frac{n_m(\text{r/min})_{\text{Freq}}}{n_m(\text{r/min})_{\text{DC}}}. \quad (28)$$

The overall improvement obtained by the proposed RI when compared to the conventional PI is clearly confirmed in time and frequency domains, as well as with the numerical computed values from the FFTs as given in Table III. Also, from Table III, it can be concluded that the cogging disturbance component (comp.) is perfectly rejected for all speeds by the proposed RI

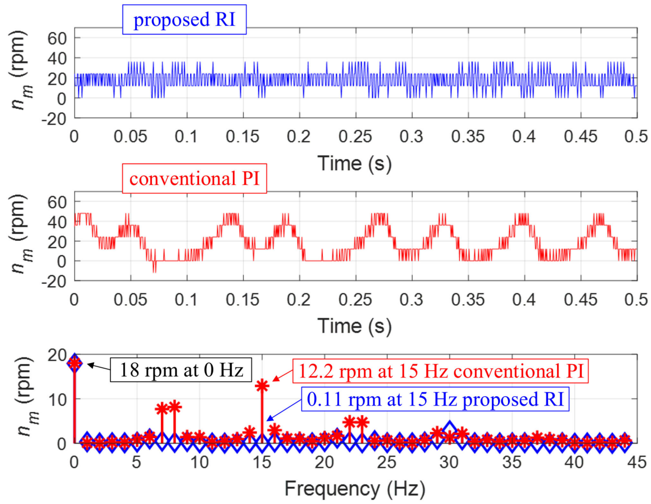

Fig. 11. Tracking 18 r/min comparison. Time waveforms and their FFTs (proposed RI in blue and conventional PI in red).

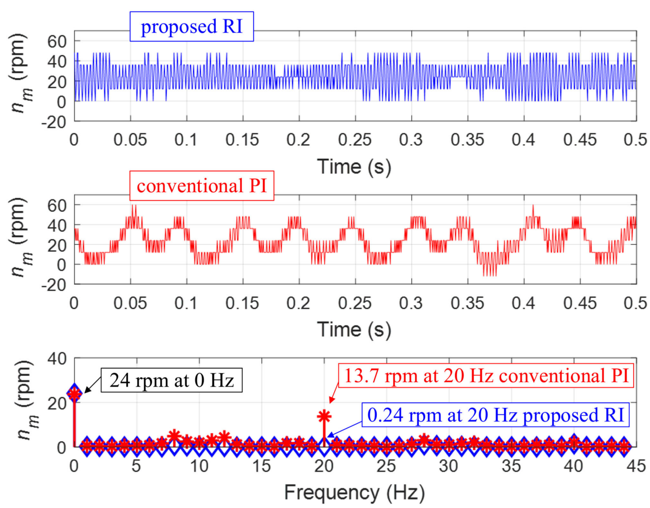

Fig. 12. Tracking 24 r/min comparison. Time waveforms and their FFTs (proposed RI in blue and conventional PI in red).

TABLE III
TOTAL HARMONIC DISTORTION (THD) AND COGGING COMPARISONS OF FIGURES 9–12 FFTS FOR THE PROPOSED RI VERSUS THE CONVENTIONAL PI

	THD			Cogging			
Speed rpm	RI -	PI -	Att. %	Comp. Hz	RI rpm	PI rpm	Att. dB
6	5.57	9.97	178.9	5	0.18	10.02	34.91
12	2.15	6.48	301.4	10	0.27	13.45	33.94
18	0.81	3.97	490.1	15	0.11	12.20	40.89
24	0.49	2.58	526.5	20	0.24	13.70	35.13

and the calculated attenuation (Att.) matches reasonably well with the theoretical −39 dB of Fig. 5.

However, despite the overall better performance of the proposed RI, the second harmonic component is not always of lower value. Actually, in Figs. 10 and 11, the second harmonic components, which lie in 20 and 30 Hz, respectively, are of larger value in the RI. A possible solution would be to include another resonant controller tuned to cancel it.

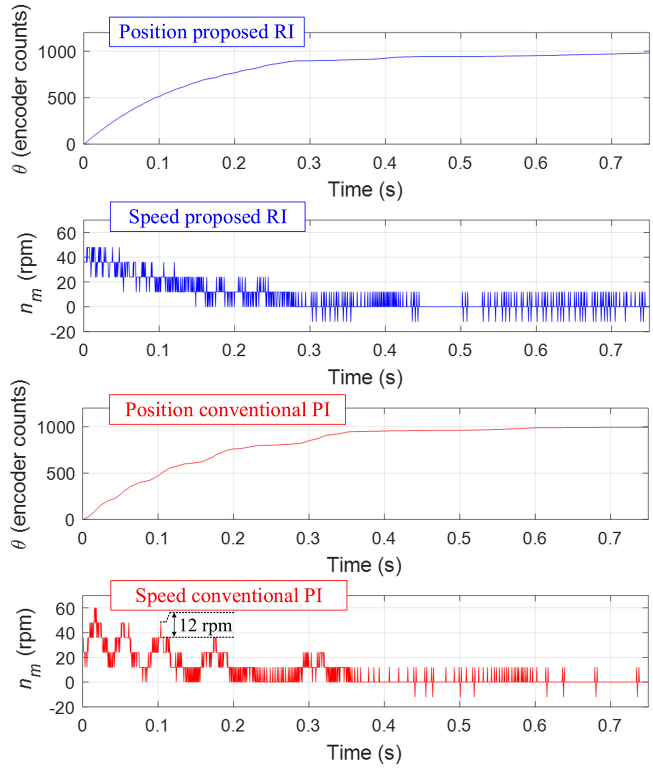

Fig. 13. Position closed-loop step response at 1000 in encoder counts comparison. Position and speed time responses (proposed RI in blue and conventional PI in red).

C. Position Loop

Position control is somehow the most difficult test to be handled by the RI controller, since the speed is constantly changing and so does the resonant controller. Resonant coefficients cannot be updated with the measured speed, since it would be a loop inside a loop, and instead they are updated with the reference value (prefiltered to emulate the closed-loop dynamics). Despite all the previous hindrances, the results are of superior performance for the proposed RI, as it can bee seen in Fig. 13, where there has been a step change from −1000 to +1000 encoder counts and the last 1000 counts are illustrated. The 12 r/min encoder resolution clearly further perturbs the position closed-loop. Actually, the measured position is not in its final value yet, since with the adopted value of $Cp(z) = 2$, speed reference figures when approaching the position reference are so tiny. For example, when the error in encoder counts E_θ(counts) is equal to 10, the speed reference, according to (29), would be as tiny as 0.12 r/min

$$\begin{aligned} n_m^*(\text{r/min}) &= E_\theta(\text{counts}) \cdot \frac{1(\text{rev})}{10^4(\text{counts})} \cdot \frac{60(s)}{1(\text{min})} \cdot Cp \\ &= E_\theta(\text{counts}) \cdot 120 \cdot 10^{-4}. \end{aligned} \tag{29}$$

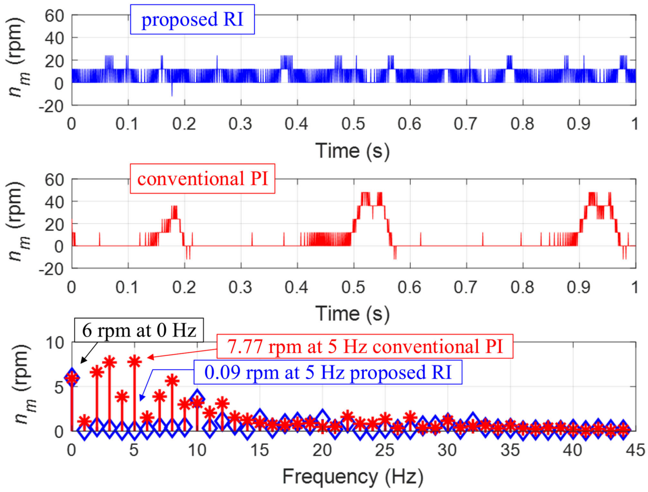

Fig. 14. Tracking 6 r/min against inertia variation ($J = 0.05 \cdot 10^{-3}$ kg·m^2). Time waveforms and their FFTs (proposed RI in blue and conventional PI in red).

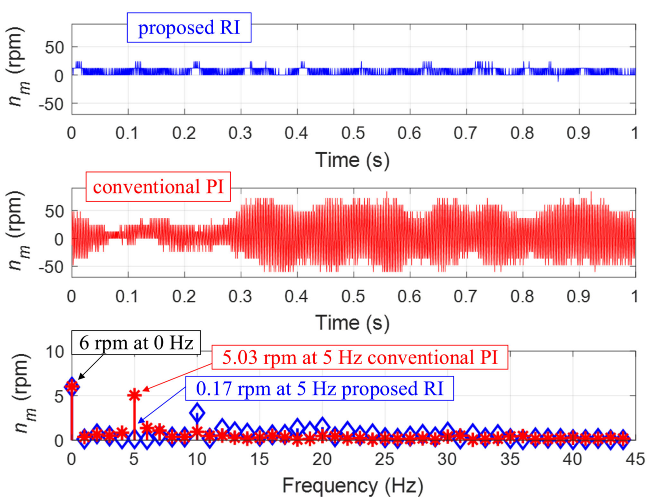

Fig. 15. Tracking 6 r/min against inertia variation ($J = 0.5 \cdot 10^{-3}$ kg·m^2). Time waveforms and their FFTs (proposed RI in blue and conventional PI in red).

Such low value of $Cp(z)$ has been chosen in order to get an exponential position waveform with a slow time constant, and therefore, maximize the duration of both tiny reference speeds and unwanted cogging torque disturbances.

D. Robustness Against Inertia (J) Variation

In PMHSM drives the variation of the inertia (J) is likely to happen as in other servo motor applications. Several tests with the original rig (i.e., with the same inertia) but tuning both (proposed RI and conventional PI) controllers with different inertia values have been undertaken. Extreme J values, ($0.05 \cdot 10^{-3}$ and $0.5 \cdot 10^{-3}$ in kg·m^2) are illustrated in Figs. 14 and 15, respectively, and it can be concluded that robustness against inertia variation is much superior for the proposed RI than the conventional PI. Also, cogging reduction in the proposed RI is maintained and the tracking performance of the conventional PI is deeply deteriorated, while the proposed RI is not.

TABLE IV
PARAMETERS OF THE SECOND TEST BENCH

PMHSM (reference SY86STH118-6004B)		
Parameter	**Value**	**Units**
Step angle / Phase no. / N_r	1.8 / 2 / 50	° / — / —
R / L / Voltage constant (KE)	4.0 / 0.5 / 0.916	Ω / mH / V·s·rad^{-1}
Holding / Detent Torques	8.0 / not specified	Nm / Nm
Test bench		
Parameter	**Value**	**Units**
Inertia / Friction	$0.64 \cdot 10^{-3}$ / $54.2 \cdot 10^{-3}$	kg·m^2 / Nm·s·rad^{-1}
Sampling period (T)	$500 \cdot 10^{-6}$	s
Encoder pulses / Speed resol.	4·1000 / 30	counts·rev^{-1} / rpm

TABLE V
TOTAL HARMONIC DISTORTION (THD) AND COGGING COMPARISONS FOR THE PROPOSED RI VERSUS THE CONVENTIONAL PI FOR THE SECOND PMHSM

	THD			Cogging			
Speed rpm	RI -	PI -	Att. %	Comp. Hz	RI rpm	PI rpm	Att. dB
6	4.83	11.64	240.1	5	0.39	10.78	28.83
12	1.22	6.35	520.5	10	0.23	17.11	37.43
18	0.38	5.88	1547	15	0.09	19.08	46.52
24	0.22	2.08	945.4	20	0.09	22.80	48.07

VII. EXTENSION TO OTHER PMHSMs

In order to validate the proposed resonant controller with other PMHSMs, the same experimental speed and position tests have been repeated with a second PMHSM detailed in Table IV.

The same controller parameters as in Table II are used, with two exceptions: 1) the damping of the poles ζ_p is equal to 0.001, which implies an additional -20 dB at the resonance peak; and 2) the overall controller RI gain K has been set to 0.08 in order to meet the equivalent performance in the sensitivity function. Moreover, the ripple in the speed measurement is equal to 30 r/min (instead of 12) due to the new 4000 encoder pulses as indicated in Table IV.

Table V summarizes the speed results, while the position test is illustrated in Fig. 16. Higher levels of cogging attenuation (48.07 dB at 24 r/min) are achieved, which is consistent with the fact of having a larger damping value ($\zeta_p = 0.001$) of the resonant poles. It can be concluded that the proposed resonant controller has superior performance compared to the conventional PI for this second PMHSM.

VIII. CONCLUSION

A different approach to minimize the cogging torque in hybrid stepper machines, based on the use of resonant controllers, was proposed in this paper. The proposed controller, composed of a speed-adaptive resonant controller as well as an integrator, proved to be a good choice for mitigating or eventually rejecting the cogging torque and minimizing its undesired consequences.

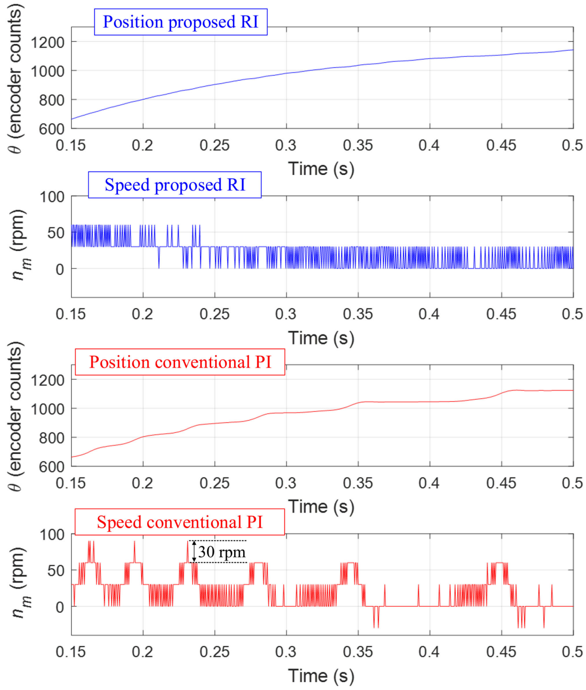

Fig. 16. Position closed-loop step response at 1200 in encoder counts comparison for the second PMHSM. Position and speed time responses (proposed RI in blue and conventional PI in red).

A Z-domain comprehensive analysis based on the pole-zero placement technique and sensitivity function frequency response was detailed. The challenge of adapting the resonance frequency of the resonant controller in accordance with the cogging disturbance-torque-frequency, which is a multiple of the mechanical speed, was achieved.

A four-quadrant workbench composed of an off-the-shelf PMHSM attached to dc motor was built and a comparison of the proposed controller against a conventional PI based one, both implemented in the same 32-bit floating-point DSC, was carried out. As a result, a set of tracking and disturbance rejection experimental tests was obtained, which illustrates the superior performance not only in low and zero closed-loop speed control, but also in closed-loop position control as well as against inertia parameter variations.

Despite the proposed control algorithm being experimentally tested for two different PMHSMs, the authors believe that this methodology can be used with success in other ac machines, where the relation between the cogging torque and mechanical frequencies are much lower than 50 (as in the PMHSMs), and therefore, it would be more favourable.

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